Project Manager: A Team Project and Task Tracking System

Software Engineering Course Project – Phase 1 Project Proposal

Team Members: Ethan Van Drunen and Diego Chavez Espinosa

Purdue University Northwest – September 27, 2026

1. Problem Statement

When a manager is running a large team or several projects at once, it is hard to keep everyone on track and hit the due dates. Right now most of this is done by hand. Tasks get handed out in meetings, emails, and texts, progress is kept in spreadsheets or on whiteboards, and the only way for a manager to really know how a project is going is to check in with each person or ask them to write a status report.

Why this needs a software solution

The information a manager needs (who is working on what, when it is due, and whether it is done) already exists, but it is spread out across people and tools. Gathering it by hand gets harder with every new project or team member. If tasks, due dates, and progress are all kept in one system, the manager can see everything in one place without having to go ask for it.

Current pain points

- **Time wasted on check-ins:** Managers spend a lot of time checking in on each project and each person just to find out where things stand.
- **Missed deadlines found too late:** Since progress is not visible day to day, a task that is behind usually is not noticed until it is already late.
- **Extra reporting for team members:** Team members have to write progress reports on top of doing the actual work.
- **Uneven workloads:** Without seeing who has how many tasks, some people end up overloaded while others have room for more.
- **Lost or unclear tasks:** Tasks given out in person or buried in long email threads are easy to forget or misunderstand.

Potential impact

Our system will let managers see all of their projects and team members on one screen without checking in on everyone. When team members update their tasks, their progress shows up for the manager automatically, so no separate reports are needed. Late tasks are flagged right away, and a workload view helps managers split up work fairly and set realistic due dates. The result should be fewer missed deadlines, less time spent in status meetings, and a more even workload for the team.

2. Target Users and Stakeholders

A user is someone who uses the software directly. A stakeholder is someone who is affected by it or depends on it.

Role or group	Type	Main need or concern
Managers / project leads	Primary user	See how every project and team member is doing without constant check-ins; create projects, give out tasks, and set due dates.
Team members / employees	Primary user	A clear list of their own tasks and due dates; their progress is shown to the manager without writing reports.
System administrator	Secondary user	Set up and remove user accounts and decide who is a manager and who is a team member.
Company leadership	Stakeholder	Projects finished on time and workloads that are fair to employees.
Clients	Stakeholder	Count on the project due dates that the system helps managers meet.
Development team (Ethan and Diego)	Stakeholder	A project small enough to split between two people and finish this semester.

System scope

What we will build: create projects, create and assign tasks, set due dates, update task progress, show how much of each project is done and how much each person contributed, flag late tasks, show each person's workload, and a login for managers and team members.

What we will not build: time clock or payroll, file sharing, built-in chat or video calls, advanced scheduling charts, email or text message alerts (alerts will only show inside the app), a phone app, and connections to other apps like Slack or Google Calendar.

Assumptions and constraints

- **Platform:** The system will be a website that works in any normal web browser on a computer or phone.
- **Team size:** It has to work for both small teams and large companies, so it cannot be built around one team size.
- **Login:** Everyone has to log in to use it, and only managers can create projects, give out tasks, or change due dates.
- **Ethics and privacy:** Workload and progress info should be used to set fair due dates and workloads, not to punish people. A person's workload is only visible to them and their manager.

3. Functional Requirements

These describe what the system will do. Each one includes a simple test to check that it works.

ID	Requirement	User	How we will check it
FR-1	The system shall let a manager create a project with a name, description, start date, and due date.	Manager	Create a project and check that all four details are saved and show up in the project list.
FR-2	The system shall let a manager create a task in a project and assign it to one or more team members.	Manager	Assign a task to two people and check that it shows up on both of their task lists.
FR-3	The system shall let a manager set and change the due date for each task and project.	Manager	Set a due date, change it, and check that the new date shows for both the manager and the team member.
FR-4	The system shall let a team member mark their task as Not Started, In Progress, or Completed.	Team member	Change a task to each of the three options and check that the manager sees each change.
FR-5	The system shall show what percent of each project is done and what percent of the finished work each person did.	Manager	Finish 1 out of 4 tasks and check that the project shows 25% done.
FR-6	The system shall show the manager a main screen listing all their projects, how much of each is done, how many tasks are late, and the next due date.	Manager	Set up 3 projects with known numbers and check that the main screen shows the right numbers.
FR-7	The system shall mark a task as late if it is not finished by its due date and show an alert to the team member and manager.	Manager, team member	Make a task that was due yesterday and is not finished; check that it is marked late and both people get an alert.
FR-8	The system shall show how many unfinished tasks each team member has.	Manager	Give one person 3 tasks and another person 1; check that it shows 3 and 1.
FR-9	The system shall require a username and password to log in, and only give managers access to manager features.	All users	Try to open the app without logging in (should be blocked); log in as a team member and check that manager features are hidden.

4. Non-Functional Requirements

These describe how well the system needs to work. Each one has a number or clear goal so it can be tested.

ID	Category	Requirement	How we will check it
NFR-1	Speed	The manager's main screen shall load in 2 seconds or less, even with 50 projects and 1,000 tasks in the system.	Add 50 projects and 1,000 tasks, load the main screen 20 times, and time it. Almost every load should take 2 seconds or less.
NFR-2	Speed	When a team member updates a task, the change shall show up on the manager's screen within 3 seconds.	Update a task as a team member and time how long it takes to show up for the manager.
NFR-3	Security	Passwords shall be scrambled (encrypted) before they are saved, and an account shall be locked for 15 minutes after 5 wrong password tries.	Look at the saved data and check that no password can be read. Type the wrong password 5 times and check that the account locks.
NFR-4	Ease of use	A new manager shall be able to create a project and assign a task in under 5 minutes without any training.	Have 5 people who have never used the system try it. At least 4 of them should finish in under 5 minutes.
NFR-5	Reliability	The system shall be up and working at least 99% of the time and save a backup copy of all data once a day.	Track how often the system is down over one month. Delete test data and check that it can be brought back from the backup.
NFR-6	Handling more users	At least 200 people shall be able to use the system at the same time without the main screen taking longer than 2 seconds to load.	Use a testing tool to act like 200 people using the system at once, and time how long the main screen takes to load.
NFR-7	Privacy	Team members shall only be able to see projects they are part of. Only the person and their manager can see that person's workload.	Log in as someone on Project A and try to open Project B or another person's workload. Both should be blocked.